

CHEMISTRY BOARD QUESTION PAPER: FEBRUARY 2025**Time: 3 Hrs. Max. Marks: 70****General Instructions:**

1. The question paper is divided into four sections.
2. Section A: Q.1 contains Ten multiple choice type questions carrying One mark each.
Only the first attempt will be considered for evaluation.
Q.2 contains Eight very short answer type questions carrying One mark each.
3. Section B: Q.3 to Q.14 are Twelve short answer type questions carrying Two marks each. (Attempt any Eight.)
4. Section C: Q.15 to Q.26 are Twelve short answer type questions carrying Three marks each. (Attempt any Eight.)
5. Section D: Q.27 to Q.31 are Five long answer type questions carrying Four marks each. (Attempt any Three.)
6. Use of log table is allowed. Use of calculator is not allowed.
7. Figures to the right indicate full marks.
8. Given:
 - i. $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
 - ii. Atomic mass of Na = 23
 - iii. K_f for water = $1.86 \text{ K kg mol}^{-1}$
 - iv. $IF = 96500 \text{ C}$
 - v. $N_A = 6.022 \times 10^{23}$

SECTION – A**Q.1. Select and write the correct answers for the following multiple choice type of questions: [10]**

- i. Schottky defect is NOT observed in
(A) NaCl (B) KCl (C) AgBr (D) NiO
- ii. The freezing point of 0.1 m aqueous solution of urea, if K_f for water is $1.86 \text{ K kg mol}^{-1}$ is:
(A) 1.86°C (B) -1.86°C (C) 0.186°C (D) -0.186°C
- iii. Ozone layer is depleted by
(A) NO (B) NO_2 (C) NO_3 (D) N_2O_5



iv. When excess of AgNO_3 is added to a complex, one mole of AgCl is precipitated. The formula of complex is:

- (A) $[\text{CoCl}_2(\text{NH}_3)_4]\text{Cl}$ (B) $[\text{CoCl}(\text{NH}_3)_5]\text{Cl}_2$
(C) $[\text{CoCl}_3(\text{NH}_3)_3]$ (D) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$

v. The value of Δn for oxidation of 4 mole of SO_2 to SO_3 is:

- (A) -2 (B) -1 (C) 1 (D) 4

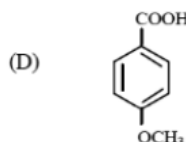
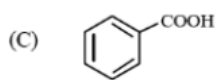
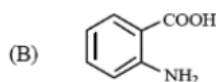
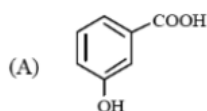
vi. One-dimensional nanostructure among the following is:

- (A) Nanoparticles (B) Nanotubes (C) Nanofilms (D) Nanorods

vii. Which formula relates degree of dissociation α and concentration of electrolyte?

- (A) $\alpha = \sqrt{K_a / c}$ (B) $\alpha = \sqrt{K_c / a}$
(C) $\alpha = \sqrt{c / K_a}$ (D) $\alpha = c / \sqrt{K_a}$

viii. The highest acidic compound among the following is:



ix. The formula used to calculate molar conductivity of an electrolyte is:

- (A) $\Lambda_m = 1000 \kappa / c$
(B) $\Lambda_m = 1000 \Lambda / \kappa$
(C) $\Lambda_m = \kappa / 1000 c$
(D) $\Lambda_m = \kappa \times 1000 / \Lambda c$

x. Which of the following is a secondary amine?

- (A) Cyclohexylamine (B) Isopropylamine
(C) Diphenylamine (D) N,N-Dimethyl ethylamine

Q.2. Answer the following questions: [8]

i. Write the structural formula of N,N-dimethylethylamine.

ii. Write the reagents used for the reduction of carbonyl group in Clemmensen's reduction.



- iii. Write the IUPAC name of isoprene.
- iv. For reaction $A \rightarrow \text{Product}$, rate = $k[A]^n$. What is effect of increase in concentration of A on rate if $n < 0$?
- v. What is molality of aqueous KBr solution with freezing point -3.72°C ? ($K_f = 1.86 \text{ K kg mol}^{-1}$)
- vi. Write balanced equation when excess NH_3 reacts with chlorine gas.
- vii. Write number of donor atoms present in EDTA during complex formation.
- viii. Write names of metal elements in brass alloy.

SECTION – B

Attempt any EIGHT of the following: [16]

- Q.3. Derive relation between half-life and rate constant for first-order reaction. [2]
- Q.4. (a) State Henry's law. (b) Define osmotic pressure. [2]
- Q.5. Write differences between lanthanoids and actinoids. [2]
- Q.6. Write anomalous behaviour of oxygen w.r.t: (i) Oxidation state (ii) Nature of hydrides. [2]
- Q.7. What is action of: (i) Liquid bromine in acetic acid on anisole (ii) Sodium–amalgam on sodium acetate? [2]
- Q.8. Calculate work done if volume of reactant decreases from 8 dm^3 to 4 dm^3 against 34 bar pressure. [2]
- Q.9. Explain ionisation isomers with suitable example. [2]
- Q.10. Write preparation of glucose from sucrose. [2]
- Q.11. How many coulombs of electricity are required to produce 1 g sodium metal from Na^+ ? [2]
- Q.12. Write structural formula and IUPAC name of alcohol $\text{C}_4\text{H}_{10}\text{O}$ which does NOT undergo oxidation. [2]
- Q.13. Identify A and B:
 $\text{CH}_3-\text{CH}=\text{CH}_2 \xrightarrow{\text{HBr/Peroxide}} \text{A} \xrightarrow{\text{alc.KOH}} \text{B}$ [2]
- Q.14. Write preparation of: (i) Acetaldehyde (Rosenmund) (ii) Benzaldehyde (Gattermann–Koch). [2]

SECTION – C

Attempt any EIGHT: [24]

- Q.15. Write general electronic configuration of 3d series. Draw structures of H_2SO_4 and thiosulphuric acid. [3]
- Q.16. Define conjugate acid–base pair. Calculate pH of NaOH ($2 \times 10^{-4} \text{ mol dm}^{-3}$). [3]
- Q.17. What is atom economy? Explain two applications of nanomaterials. [3]



- Q.18. What is peptide bond? How is it formed? Give name & formula of reagent used for Hofmann rearrangement. [3]
- Q.19. Write reactions for the action of the following on phenol: (i) Nitrating mixture (ii) Zinc dust. [3]
- Q.20. (a) Why phosphorus pentachloride does not form ethyl methyl ether?
(b) Write formula & calculate EAN of $[\text{Co}(\text{NH}_3)_6]^{3+}$. [3]
- Q.21. (a) Calculate S.O.M of M^{3+} ion. ($\mu = 2.6 \text{ BM}$)
(b) Write electronic configuration of Gd. [3]
- Q.22. (a) Write two reactions in which Fe^{3+} converts Fe_2O_3 into Fe in blast furnace.
(b) Chemical equations for:
(i) Carbylamine reaction (ii) Hoffmann Bromamide (acetamide). [3]
- Q.23. (a) Explain Cannizzaro's reaction.
(b) Write reaction for cyclohexene $\rightarrow$ adipic acid. [3]
- Q.24. (a) Define zero-order reaction.
(b) A reaction takes place in two steps:
 $\text{NO}_2 + \text{Cl}_2 \rightarrow \text{NO}_2\text{Cl}$
 $\text{NO}_2\text{Cl} + \text{NO}_2 \rightarrow 2\text{NO}_2\text{Cl}$
Write overall reaction & identify intermediate. [3]
- Q.25. ΔH formation of ethane gas = -84.4 kJ at 300 K . Calculate ΔU . [3]
- Q.26. Mention three types of polymers based on intermolecular forces. Write two uses of LDPE. [3]

SECTION – D

Attempt any THREE: [12]

- Q.27. (a) An element of molar mass 97 g/mol forms a cubic unit cell with edge 405 pm . Density = 2.7 g/cm^3 . Identify type of unit cell.
(b) Derive Raoult's law equation for binary solution. [4]
- Q.28. (a) State whether entropy change is +ve or -ve in:
(i) Melting of ice (ii) Vaporisation of liquid.
(b) Explain common-ion effect with example. [4]
- Q.29. Draw neat labelled diagram of lead accumulator. Write overall reactions at anode & cathode during discharge. [4]
- Q.30. (a) Define unit cell.
(b) Which colour is shown by NaCl crystal due to F-centres?
(c) Why does fluorine show anomalous behaviour in group 17? [4]
- Q.31. (a) Write salient features of SN_1 mechanism.
(b) What is action of following reagents on trichloromethane?
(i) $\text{Br}_2 / \text{NaOH}$ (ii) Mercuric fluoride. [4]

